Cian Roche

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Education

Massachusetts Institute of Technology 2021 – Present
PhD candidate - Physics
Thesis: *"The Dark Universe Through The Strong Lens: Cluster and Galaxy-Scale Probes of the Dark Sector"*
Advisors: Prof. Michael McDonald, Prof. Mark Vogelsberger

Universität Tübingen 2019 – 2021
MSc Mathematical Physics 1.1 GPA[†]
Thesis: *"Exact parallel waves in general relativity"*
Advisor: Prof. Carla Cederbaum

University College Cork 2015 – 2019
Bachelor of Science: Physics, 1st class honours

Honors and Awards

MIT Physics Henry W Kendall Teaching Fellowship 2026
DAAD Graduate Program Full Scholarship 2019
College Scholar - University College Cork 2019
Max Planck Institute for Plasma Physics Research Scholarship 2018
Naughton Research Fellowship (one of 8 in Ireland) 2017
Naughton Foundation Scholarship (one of 29 in Ireland) 2015
All-Ireland Scholarship (one of 125 in Ireland; declined) 2015

Teaching

Massachusetts Institute of Technology
Teaching Assistant, Experimental Physics II (8.14) ..... Spring 2026
Teaching Assistant, Experimental Physics I (8.13) ..... Fall 2024
Teaching Assistant, Experimental Physics I (8.13) ..... Spring 2024
Teaching Assistant, Experimental Physics I (8.13) ..... Fall 2023
Teaching Assistant, Experimental Physics I (8.13) ..... Fall 2022
Instructor - Gaia DR3 Hackathon ..... Jun 2022

University College Cork
Teaching Assistant, Introductory Physics II ..... Spring 2018
Leader, Peer Assisted Learning Scheme ..... 2016 – 2018

Publications

First Author or Significant Contribution

4. Tran, V., Shen, X., Vogelsberger, M., Gilman, D., O'Niell, S., **Roche, C.**, Zier, O., Gao, J. 2024; *"Novel Density Profile for Isothermal Cores of Dark Matter Halos"*, Phys. Rev. D 113, 063024, arXiv:2411.11945
3. **Roche, C.**, McDonald M., Vogelsberger, M., Shen, X., Springel, V., Hernquist, L., Pakmor, R., Bose, S., Kannan, R. 2024; *"Brightest Cluster Galaxy Offsets in Cold Dark Matter"*, OJA, astro-ph.GA 7, arXiv:2402.00928
2. **Roche, C.**, Necib, L., Lin, T., Ou, X., Nguyen, T. 2024; *"The Escape Velocity Profile of the Milky Way from Gaia DR3"*, ApJ, 972, 70, arXiv:2402.00108
1. **Roche, C.**, Aazami, A.B. & Cederbaum, C. 2023; *"Exact parallel waves in general relativity"* Gen Relativ Gravit 55, 40, doi:10.1007/s10714-023-03083-x

[†]on a scale of 1-5, where 1 is the highest grade.

Nth Author

3. Rose J. C., et al. (incl. **Roche, C.**) 2025; *"The DREAMS Project: A New Suite of 1,024 Simulations to Contextualize the Milky Way and Assess Physics Uncertainties"*, arXiv:2512.00148
2. Nguyen, T., et al. (incl. **Roche, C.**) 2024; *"How DREAMS are made: Emulating Satellite Galaxy and Subhalo Populations with Diffusion Models and Point Clouds"*, ApJ, 997, 336, arXiv:2409.02980
1. Rose J. C., et al. (incl. **Roche, C.**) 2024; *"Introducing the DREAMS Project: DaRk mattEr and Astrophysics with Machine learning and Simulations"*, ApJ, 982, 68, arXiv:2405.00766

Non-Refereed

1. **Roche, C.** & Garnavich, P. 2020; "*Testing Progenitor Models Using the Late-Time Light Curve of Supernova 1992A*" Res. Notes AAS 4 207, doi:10.3847/2515-5172/abcb05

Presentations

Invited (Colloquia/Seminars)

Talk	"*Light Cone Strong Lensing in Cosmological Hydrodynamical Simulations*" MIT Kavli Institute for Astrophysics and Space Research	Mar 2026
Talk	"*Understanding Dark Matter with Simulated Strong Lensing*" Yale University	Sep 2025
Talk	"*Dark Matter Self-Interaction via Galaxy Cluster Dynamics*" Harvard University	Feb 2024
Talk	"*Galaxies Lacking Dark Matter Produced in a Cosmological Simulation*" MIT Kavli Institute for Astrophysics and Space Research	Apr 2022
Lecture	"*Waves in General Relativity*" Mathematical Relativity Class – University of Tübingen	Jul 2021
Talk	"*Exact parallel waves in general relativity*" Mathematical Physics Colloquium – University of Tübingen	Jun 2021
Talk	"*How to Stop a Hurricane: Complexity in Physics*" "Blow Your Mind Week" 2020 – University College Cork	Oct 2020
Talk	"*Hermiticity of Young Operators*" Young Tableaux Seminar Series – Universität Tübingen	Jun 2020

Contributed

Talk	"*Consistent Strong Lensing Predictions from Cosmological Simulations*" Scaling up Lensing Workshop – University of Liège	Oct 2025
Poster	"*Unveiling Dark Matter: Testing CDM with the First End-to-End Cluster Strong Lensing Simulations*" Tracing cosmic Evolution with Galaxy Clusters V – Sesto, Italy	Jul 2025
Talk	"*Testing CDM with the First End-to-End Cluster Strong Lensing Simulations*" European Astronomical Society Meeting 2025 – University College Cork	Jun 2025

Community and Service

Astrogazers 2024–Present
Massachusetts Institute of Technology
Volunteered for general public and youth outreach events with MIT astrogazers, including Massachusetts Space Week and Cambridge Science Festival.

MIT Physics Graduate Admissions 2022/2023 Cycle, 2023/2024 Cycle
Massachusetts Institute of Technology
Served for two admissions cycles on the admissions committee for the graduate program in physics as part of the SAMC (student admissions committee members) initiative.

AstroWiki 2023 – Present
Massachusetts Institute of Technology
Created and co-maintain a set of living notes covering most of graduate level astrophysics, specifically designed to aid the next generations of graduate students when studying for the MIT astrophysics oral qualification exam. (link)

MIT Summer Research Program Admissions Committee 2022/2023 Cycle
Massachusetts Institute of Technology
Served on the admissions committee for the MIT Summer research program (MSRP), which aims in particular to provide high-quality research experiences to promising undergraduate students from underrepresented backgrounds.

MIT GAGA Initiative Co-chair 2022 – 2023 | Organising Committee 2021 – 2024
Massachusetts Institute of Technology
The Graduates Advising Graduate Admissions (GAGA) program provides the graduate student perspective to the Chair of Graduate Admissions, offers information and mentorship to underrepresented groups in physics and advises on improvements to the admissions process.

MIT PhysGAAP program 2021 – 2023
Massachusetts Institute of Technology
The Physics Graduate Application Assistance Program (PhysGAAP) provides guidance to prospective applicants on navigating the application process, and liases with the physics department leadership on issues of accessibility and equity in the graduate application. I have served as one of the primary organizers and have mentored 5 students through the program.

UCC Physics and Astronomy Society 2017 – 2019
University College Cork
Served as the Astronomy officer of the University College Cork Physics and Astronomy Society, organizing large astronomy outreach trips to dark sky regions in Western Ireland, and aurora borealis observation in Iceland.

Miscellaneous Volunteer Positions
Volunteer, AAS 241st Meeting Graduate School Fair 2023
Desk Czar/building management liason, MIT Kavli Institute for Astrophysics and Space Research . 2022–2025
Organizer, MIT Kavli Institute for Astrophysics and Space Research graduate student lunch 2022–2023
Member, MIT Physics Graduate Student Council ..... 2021 – 2023
Organising Committee, MIT Graduate Student Union ..... 2021 – 2023